

SplitTab - Product Requirements Document

Version 1.0 - draft for engineering

1. Overview

SplitTab is a mobile-first web application for splitting shared expenses within a small group - flatmates, a holiday party, a recurring dinner club. A member records an expense once, SplitTab keeps a running balance for everyone, and at any point a member can ask for the minimum set of payments that clears all debts in the group.

The product is deliberately narrow. It does not move money, it does not support multiple currencies in v1, and it has no social feed. Its value is that the arithmetic is always right and always visible.

2. Users

Member - anyone who has joined a group. Can add expenses, view balances, and settle up.

Group owner - the member who created the group. Can invite and remove members and rename the group. There is exactly one owner and ownership can be transferred.

3. Functional requirements

FR-1 Accounts. A user signs up with email and password, and signs in to receive a session. Passwords are stored hashed. Sessions expire after 30 days.

FR-2 Groups. A signed-in user can create a group with a name and a base currency. The creator becomes the owner and the first member.

FR-3 Invitations. An owner can invite a member by email. The invitation produces a single-use link valid for 7 days. Accepting it adds the user to the group.

FR-4 Expenses. A member records an expense with a description, an amount, a date, a payer, and the set of members it is shared between. An expense may be split equally, by exact amounts, or by percentage shares. Splits must sum to the expense total; a split that does not sum is rejected with an explanation of the difference.

FR-5 Editing. Any member of the group can edit or delete an expense. Every change is recorded in an immutable audit trail showing who changed what and when. Balances recompute after any edit.

FR-6 Balances. Each member sees, for every group, what they owe and what they are owed, per counterparty. The sum of all balances in a group is always zero.

FR-7 Settlement. A member can ask for a settlement plan: the minimum number of payments that brings every balance to zero. Recording a settlement payment updates the balances the same way an expense does.

FR-8 Notifications. A member receives an email when they are added to a group, when an expense involving them is recorded, and a weekly digest of their open balances. Each of the three is individually switchable in the member's settings.

FR-9 Export. A member can export a group's expense history as CSV, covering an optional date range, with one row per expense and one column per member's share.

FR-10 Activity view. A group has a reverse-chronological activity feed of expenses, edits, settlements, and membership changes, paged at 50 entries.

4. Non-functional requirements

NFR-1 Money is never stored or computed in floating point. All amounts are integer minor units, and every split must reconcile to the cent.

NFR-2 A group of 20 members with 5,000 expenses loads its balance view in under 500 ms at the 95th percentile.

NFR-3 Every write is authorised against group membership. A user must not be able to read or alter a group they do not belong to, including by guessing an id.

NFR-4 The settlement algorithm is covered by property-based tests asserting that the resulting balances are all zero and that no member pays more than they owed.

5. Out of scope for v1

Payment processing, multi-currency groups, receipt image OCR, recurring expenses, and native mobile applications.